

CHE 312 HW #4

Saturday, September 27, 2025

5:35 PM

Q1. a) $q_1'' = \frac{k\Delta T}{L}$

$$q_1'' = \frac{237 \text{ W/m}\cdot\text{K} (310-290) \text{ K}}{400 \times 10^{-6} \text{ m}}$$

$$q_1'' = 1.185 \times 10^7 \frac{\text{W}}{\text{m}^2} \quad @ \quad L = 400 \mu\text{m}$$

$$q_2'' = \frac{273 \text{ W/m}\cdot\text{K} (310-290) \text{ K}}{400 \times 10^{-9} \text{ m}}$$

$$q_2'' = 1.185 \times 10^{10} \frac{\text{W}}{\text{m}^2} \quad @ \quad L = 400 \text{ nm}$$

b) $q_1'' = \frac{0.0263 (310-290)}{400 \times 10^{-6} \text{ m}}$

$$q_1'' = 1315 \frac{\text{W}}{\text{m}^2}$$

$$q_2'' = \frac{0.0263 (310-290)}{400 \times 10^{-9} \text{ m}}$$

$$q_2'' = 1.135 \times 10^6 \frac{\text{W}}{\text{m}^2}$$

c) $R = \frac{2\delta}{k_{\text{Al}}} + \frac{L-2\delta}{k_{\text{air}}}, \quad q'' = \frac{\Delta T}{R}$

$$R_1 = \frac{2(40 \times 10^{-6} \text{ m})}{237 \text{ W/m}\cdot\text{K}} + \frac{(400 \times 10^{-6} \text{ m}) - 2(40 \times 10^{-6} \text{ m})}{0.0263 \text{ W/m}\cdot\text{K}}$$

$$= 0.01217 \text{ W/m}^2\cdot\text{K}$$

$$q_1'' = \frac{(310-290) \text{ K}}{0.01217 \text{ W/m}^2\cdot\text{K}}$$

$$q_1'' = 1644 \frac{\text{W}}{\text{m}^2}$$

$$R_2 = \frac{2(40 \times 10^{-9} \text{ m})}{237 \text{ W/m}\cdot\text{K}} + \frac{(400 \times 10^{-9} \text{ m}) - 2(40 \times 10^{-9} \text{ m})}{0.0263 \text{ W/m}\cdot\text{K}}$$

$$K_2 = \frac{-1.10 \times 10^{-5} \text{ W/m} \cdot \text{K}}{237 \text{ W/m} \cdot \text{K}} + \frac{1.10 \times 10^{-5} \text{ W/m} \cdot \text{K}}{0.0263 \text{ W/m} \cdot \text{K}}$$

$$= 1.217 \times 10^{-5} \text{ W/m}^2 \cdot \text{K}$$

$$q_2'' = \frac{(310 - 290) \text{ K}}{1.217 \times 10^{-5} \text{ W/m}^2 \cdot \text{K}}$$

$$q_2'' = 1.644 \times 10^6 \frac{\text{W}}{\text{m}^2}$$

Q2. $Q = 4\pi k \frac{T_1 - T_2}{\frac{1}{r_1} - \frac{1}{r_2}}$

$$\Rightarrow k = \frac{Q \left(\frac{1}{r_1} - \frac{1}{r_2} \right)}{4\pi (T_1 - T_2)}$$

$$k = \frac{60 \text{ W} \left(\frac{1}{0.1 \text{ m}} - \frac{1}{0.15 \text{ m}} \right)}{4\pi (200 - 150) \text{ K}}$$

$$k = 0.106 \text{ W/m} \cdot \text{K}$$

Q3. Perpendicular:

$$R_1 = \frac{\text{thickness}}{k_i A}$$

$$R_1 = 5 \left(\frac{t}{k_c A} + \frac{t}{k_g A} \right) = \frac{5t}{A} \left(\frac{1}{k_c} + \frac{1}{k_g} \right)$$

$$G_1 = \frac{1}{R_1} = \frac{A}{5t} \frac{1}{\left(\frac{1}{k_c} + \frac{1}{k_g} \right)}$$

Parallel:

$$G_2 = 5t (k_c + k_g)$$

Ratio:

$$\frac{G_2}{G_1} = \frac{5t (k_c + k_g)}{\frac{A}{5t} \left(\frac{1}{k_c} + \frac{1}{k_g} \right)}$$

$$= \frac{25t^2}{A} (k_c + k_g) \left(\frac{1}{k_c} + \frac{1}{k_g} \right)$$

if $L = 1 \text{ in}$, $A = 1 \text{ in}^2$ and $t = 0.1 \text{ in}$

if $L = 1 \text{ in}$, $A = 1 \text{ in}^2$ and $t = 0.1 \text{ in}$

$$\frac{G_2}{G_1} = 0.25 \frac{(k_c + k_g)^2}{k_c k_g} \text{ always } > 0$$

$\Rightarrow$ therefore, parallel plates will conduct heat better